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Financial Engineering

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0. Introduction

The objective of this project is to implement and analyze some quantitative tools for derivative pricing and risk management. The study is divided into three main thematic blocks:

- **Foundational Pricing and Convergence (a-c):** Implementation of the Black-76 closed-form solution, the Cox-Ross-Rubinstein (CRR) binomial tree, and Monte Carlo (MC) simulations for a European Call option. This phase focuses on selecting optimal simulation parameters (M) and verifying theoretical convergence rates ($1/M$ for trees and $1/\sqrt{M}$ for MC).
- **Barrier Options and Sensitivities (d-f):** Transitioning to more complex structures by pricing "Up-and-Out" options with both European and American barriers. This includes a comparative analysis of Greeks (Delta and Vega) to assess the impact of barrier constraints on portfolio risk.
- **Facultative Extensions (g-i):** Exploration of variance reduction techniques (Antithetic Variables) and the pricing of Bermudan options, where the influence of varying dividend yields on early exercise features is evaluated.

By comparing these different methodologies, the project aims to highlight the trade-offs between analytical precision, numerical stability, and computational efficiency in a realistic market setting. All models are calibrated using a specific market scenario dated February 15, 2008, assuming a 4-month maturity and an underlying equity stock with a constant dividend yield.

1. Methodology and Numerical Results

1.1. a. Pricing of a European Call Option

The objective is to price a European Call Option using three distinct methodologies based on the market data in the table below.

Parameter	Symbol	Value
Underlying Price	S_0	1.00 €
Strike Price	K	1.10 €
Time to Maturity	T	4 months (1/3 year)
Volatility	σ	21.2% p.a.
Risk-free Rate	r	2.5%
Dividend Yield	q	2%

Table 1: Market Data for European Call Option Pricing

1.1.1 Black-Scholes Model (Analytical)

The benchmark price was calculated using the `blkprice` function, which implements the Black-76 model for forward prices. We define the forward price as $F_0 = S_0 e^{(r-q)T}$.

1.1.2 Cox-Ross-Rubinstein (CRR) Tree

We implemented a binomial tree,with a initial number $M = 100$ of time-steps and up (u) and down (d) movements calibrated to the Black-Scholes implied volatility: $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$. Thus, the risk-neutral probability is $q = \frac{1-d}{u-d}$.

1.1.3 Monte-Carlo (MC) Simulation

Let $F(t,T)$ denote the forward price of the underlying at time t, and let T be the maturity of the derivative. In an arbitrage-free market, by the Fundamental Theorem of Asset Pricing there exists

an equivalent martingale measure $\mathbb{Q}$ under which the derivative price at time 0 can be written as the discounted risk-neutral expectation of its payoff at maturity.
More precisely, if the payoff at time T is $\Phi(F(T,T))$, then its present value is

$$V_0 = B(0,T)\,\mathbb{E}_0^{\mathbb{Q}}[\Phi(F(T,T)]\,,$$

where $B(0,T)$ is the discount factor from T to 0.
In our setting, under the Black model, the forward price at maturity is assumed to follow a lognormal distribution under $\mathbb{Q}$ and can be represented as

$$F(t,T) = F(0,T)\exp\left(-\frac{1}{2}\sigma^2 t + \sigma\sqrt{t}\,Z\right), \qquad Z \sim \mathcal{N}(0,1),$$

where $F(0,T)$ is the forward price observed at time 0.
Therefore, the quantity of interest is the expectation

$$\mathbb{E}_0^{\mathbb{Q}}[\Phi(F(T,T)]\,,$$

which here is approximated numerically by Monte Carlo simulation.
To this end, we generate M independent standard normal random variables $Z_1, \ldots, Z_M$, and for each of them we construct a simulated realization of the terminal forward price,

$$F(T,T)^{(m)} = F(0,T)\exp\left(-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z_m\right), \qquad m = 1, \ldots, M.$$

The corresponding Monte Carlo estimator of the derivative price is then

$$\widehat{V}_0^{MC} = B(0,T)\,\frac{1}{M}\sum_{m=1}^{M}\Phi\Big(F(T,T)^{(m)}\Big)\,.$$

In our numerical implementation, we initially set $M = 100$ as a benchmark. This number is intentionally kept low at first, before increasing the number of simulations to a level that ensures a sufficiently accurate estimate.

Pricing Method	**Option Price (€)**
Exact Closed-Form	0.0163
CRR Tree	0.0163
Monte Carlo	0.0150

Table 2: Numerical results for the European Call Option price at $S_0 = 1$ €.

Consideration on parameter M: We used the same parameter values in both the Monte Carlo simulations and the CRR model, in order to remain consistent with the assignment settings already implemented. However, the Monte Carlo estimate is based on only $M = 100$ simulated paths, which is a very small sample size in this context. By the Law of Large Numbers, the Monte Carlo estimator converges to the true expected value only as the number of approaches infinity. Therefore, with such a limited number of paths, the estimate is still subject to substantial sampling error, which explains the significant difference with respect to the CRR result.

1.2. b. Selection of M

The parameter M is selected as the first value in a predefined grid for which the numerical error remains below the prescribed tolerance $tol = 0.5 \times 10^{-4}$, chosen so as to be smaller than the bid/ask spread of 0.01%. In our implementation, the search is restricted to powers of 2, to be consistent with the framework adopted in the subsequent questions. This restriction is not materially limiting in our setting, since already with 100 time steps the CRR approximation matches the benchmark price up to the order of 10^{-4}. In the Monte Carlo case, we generate once a maximum number of sample paths and then compute the price estimators on nested subsamples of sizes 2^k. This is an application of *common random numbers* (CRN): the estimators corresponding to different values of M are built from the same underlying random draws and are therefore positively correlated. As a result, the comparison across candidate values of M is more stable. We distinguish, however, between the nature of the error in the two methods:

- **CRR Steps:** M was chosen from the set $2^{1:10}$. The selection is based on the **discretization error**, calculated as the absolute difference between the CRR and the closed-form price.
- **MC Simulations:** M was chosen from the set $2^{1:20}$. In this case, the criterion is based on the **standard deviation** of the Monte Carlo estimator ($\hat{\sigma}/\sqrt{M}$).

Method	Optimized M (Steps/Sims)	Option Price (€)
CRR Tree	128	0.0163
Monte Carlo	1,048,576	0.0163

Table 3: Optimized parameters M and resulting option prices based on the tolerance $tol = 0.5 \times 10^{-4}$.

Observation: since the value of M found for the CRR model is similar to the one used in point 1.1.2, we had already obtained previously a precise approximation of the analytical price. On the other hand, the M we retrieved for Monte Carlo is much greater than the one we used in point 1.1.3, thus the resulting estimate of the price is now excellent.

1.3. c. Error Rescaling and Convergence Analysis

We verify the theoretical convergence rates by plotting the numerical errors on a log-log scale.

- **CRR Convergence ($1/M$):** As shown in the plot generated by `PlotErrorCRR.m`, the error follows a $1/M$ trend. The observed oscillations are due to the "pinning effect" where the strike price K falls between nodes at maturity.
- **MC Convergence ($1/\sqrt{M}$):** The statistical error (standard deviation of the mean) rescales as $1/\sqrt{M}$, confirming the Central Limit Theorem. This is visually represented by the slope of -0.5 in the log-log plot.

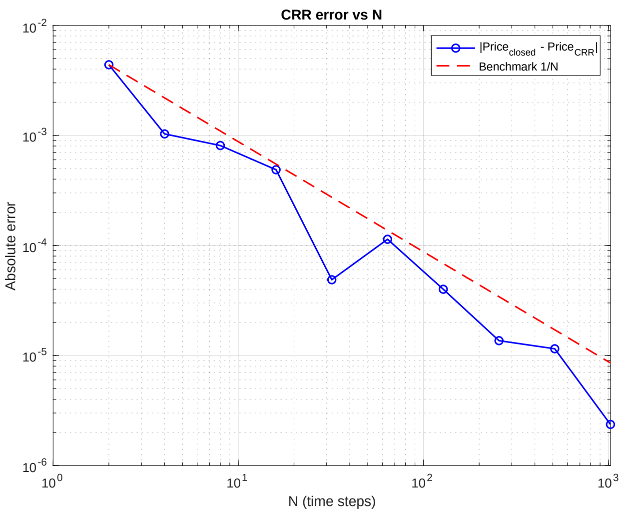

Figure 1: CRR Error vs M (Slope ≈ -1)

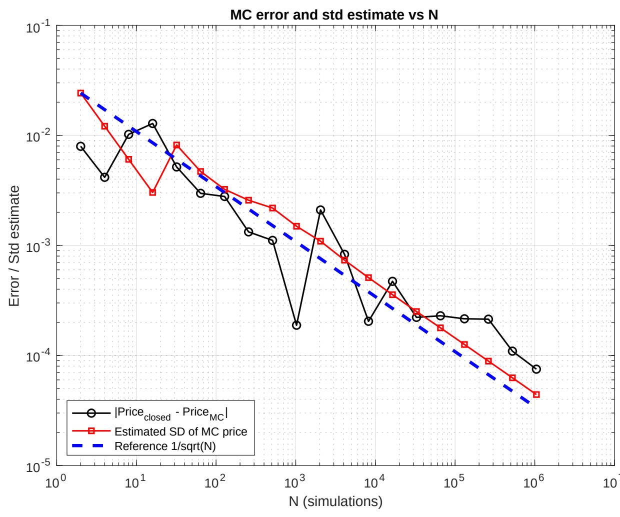

Figure 2: MC Error vs M (Slope ≈ -0.5)

1.4. d. Pricing of a European Call Option with Up-and-Out Barrier

Now, we consider a European Call Option with a European "Up-and-Out" barrier. The contract shares the same parameters as the vanilla option, with the addition of a knock-out level set at $KO = 1.40$ €. More precisely, since the barrier is monitored only at maturity (European style), the payoff at time T can be written as

$$\Phi(F(T,T)) = \max(F(T,T) - K, 0)\, \mathbf{1}_{\{F(T,T)<KO\}},$$

that is, the option pays the standard call payoff if and only if the terminal forward price observed at maturity is strictly below the barrier below the barrier ($F(T,T) < KO$).

In particular, a closed-form solution exists for this specific exotic derivative, since the payoff can be perfectly replicated using a static portfolio of vanilla instruments:

1. Long one European Call with strike K.
2. Short one European Call with strike KO.
3. Short a European Digital (Cash-or-Nothing) option with strike KO paying a fixed amount of $(KO - K)$.

Mathematically, the price of the Up-and-Out Call under the Black-76 model is given by:

$$\text{Price}_{U\&O} = \text{Call}(K) - \text{Call}(KO) - \text{Digital}(KO) \tag{1}$$

Substituting the Black-76 exact formulas and simplifying the terms, we obtain a highly symmetric and elegant closed-form equation:

$$\text{Price}_{U\&O} = e^{-rT}[F(0,T)(N(d_1(K)) - N(d_1(KO))) - K(N(d_2(K)) - N(d_2(KO)))] \tag{2}$$

The option was priced using the previously optimized parameters ($M_{CRR} = 128$ and $M_{MC} = 1,048,576$), with the additional check at maturity mentioned above. The numerical approximations from both the CRR tree and the MC simulations successfully converge to the exact analytical value provided by the replicating portfolio formula. The point-in-time results evaluated at the initial spot price $S_0 = 1$ € are summarized in Table 4.

Pricing Method	**Option Price (€)**
Exact Closed-Form	0.0154
CRR Tree ($M = 128$)	0.0154
Monte Carlo ($M = 1,048,576$)	0.0154

Table 4: Numerical results for the European Up-and-Out Call Option price at $S_0 = 1$ €.

1.5. e. Vega Sensitivity Analysis

To understand the risk profile of the barrier option, we evaluate its Vega (ν), which measures the sensitivity of the option price to changes in the implied volatility (σ). The Vega is analyzed over a range of underlying spot prices $S_0 \in [0.65, 1.45]$ €.

1.5.1 Closed-Form Vega

Exploiting the linearity of the derivative operator, the Vega of the European Up-and-Out Call, consistently with the definition adopted in the slides, is measured as the price variation corresponding to a volatility shift $\Delta\sigma = 1\%$. Hence,

$$\nu_{U\&O} \equiv \frac{\partial\,\text{Price}_{U\&O}}{\partial\sigma}\,\Delta\sigma = \left(\frac{\partial\,\text{Price}_{\text{Call}}(K)}{\partial\sigma} - \frac{\partial\,\text{Price}_{\text{Call}}(KO)}{\partial\sigma} - \frac{\partial\,\text{Price}_{\text{Digital}}(KO)}{\partial\sigma}\right)\Delta\sigma. \tag{3}$$

Equivalently, using the same convention for each component,

$$\nu_{U\&O} = \nu_{\text{Call}}(K) - \nu_{\text{Call}}(KO) - \nu_{\text{Digital}}(KO), \tag{4}$$

where each Vega is intended as the monetary price sensitivity to a 1% change in volatility.

1.5.2 Numerical Approximated Vega

The punctual numerical estimates for the Vega at $S_0 = 1$ € are based on the **Central Finite Difference** scheme. This "bumping" technique approximates the partial derivative of the option price with respect to volatility (σ) by perturbing the input parameters.

Methodology To estimate Vega, we adopt a *bump-and-revalue* finite-difference approach, in line with Glasserman's framework for Monte Carlo sensitivity estimation. If $\alpha(\sigma) = \mathbb{E}[Y(\sigma)]$ denotes the option price as a function of volatility, then Vega is the derivative $\alpha'(\sigma)$ with respect to σ. Since this derivative is not computed analytically in our numerical scheme, we approximate it by re-pricing the option at perturbed volatility levels, which is a simple method to implement but subject to a bias–variance trade-off.
We use a central-difference approximation, whose bias is of order $O(h^2)$ (instead of $O(h)$ for a forward difference), assuming sufficient smoothness of the price with respect to σ. Defining

$$\sigma_{\text{up}} = \sigma + h, \qquad \sigma_{\text{down}} = \sigma - h,$$

Vega is estimated as

$$\nu \approx \frac{\text{Price}(\sigma + h) - \text{Price}(\sigma - h)}{2h} \, \Delta\sigma,$$

with $\Delta\sigma = 1\%$, consistently with the convention adopted in the slides. Equivalently,

$$\nu \approx 0.01 \, \frac{\text{Price}(\sigma + h) - \text{Price}(\sigma - h)}{2h}.$$

The bump size is chosen as a relative perturbation,

$$h = 0.01 \, \sigma,$$

so that the volatility shift scales with the level of the parameter itself.
In the Monte Carlo implementation, the two valuations at $\sigma+h$ and $\sigma-h$ are computed using the same random seed, i.e. with *common random numbers*, in order to reduce the variance of the finite-difference estimator through positive dependence between the two bumped prices. Nevertheless, for finite h the estimator remains biased, so it should be interpreted as a numerically efficient approximation of Vega rather than an exact derivative.
The results are summarized in Table 5.

Estimation Method	**Vega**
Exact Closed-Form	0.0014
CRR Tree ($M = 128$)	0.0017
Monte Carlo ($M = 1,048,576$)	0.0014

Table 5: Point-in-time Vega estimates for the European Up-and-Out Call option at $S_0 = 1$ €.

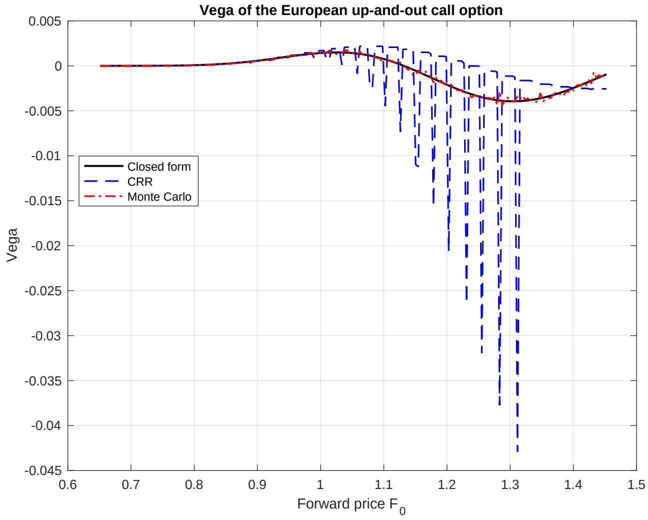

Figure 3: Vega of the European Up-and-Out Call option across different underlying prices. Notice the negative Vega near the barrier.

Consideration on Numerical Instabilities: As observed in Figure 3, the CRR model exhibits severe downward spikes as the underlying price approaches the barrier. This behavior is attributed to the **Barrier Alignment Problem**. When the volatility is bumped ($+h$), the vertical distance between nodes in the binomial tree expands, causing terminal nodes to "jump" across the continuous barrier KO. This sudden transition causes a discontinuous drop in the option price, resulting in an artificially magnified negative derivative when calculating the Vega via finite differences.

1.6. f. Pricing of a Call Option with American Up-and-Out Barrier and Greeks Sensitivity Analysis

Moreover, we extended the pricing framework to analyze a Call Option embedded with an **American-style Up-and-Out barrier**.

1.6.1 Pricing

We have already mentioned in point 1.4 the terminal Up-and-Out barrier price (where the barrier is only monitored at T), which is given by:

$$V_{Eur}(F(0,T), K, KO) = C(F(0,T), K) - C(F(0,T), KO) - B \cdot (KO - K) \cdot \Phi(d_2) \tag{5}$$

where $C(F, X)$ is the standard Black-Scholes call price for strike X, B is the discount factor, and d_2 is defined as:

$$d_2 = \frac{\ln(F(0,T)/KO) - \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \tag{6}$$

For the American-style barrier, we incorporate a **reflection term** based on the Reflection Principle to capture the effect of barrier crossing. Although the barrier is defined as continuously monitored in theory, in practice it is monitored only at discrete time points in the numerical implementation:

$$V_{Am} = V_{Eur}(F(0,T), K, KO) - \left(\frac{KO}{F(0,T)}\right)^{-1} \cdot \text{Reflected} \tag{7}$$

where the *Reflected* term is calculated by substituting the forward price $F(0,T)$ with the reflected price $F'(0,T) = \frac{KO^2}{F(0,T)}$:

$$\text{Reflected} = C(F'(0,T), K) - C(F'(0,T), KO) - B \cdot (KO - K) \cdot \Phi(d_2') \tag{8}$$

and d_2' uses the reflected geometry:

$$d_2' = \frac{\ln(KO/F'(0,T) - \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}} \tag{9}$$

The point-in-time results evaluated at the initial spot price $S_0 = 1$ € are summarized in Table 6.

Numerical Method	European Barrier (€)	American Barrier (€)
Exact Closed-Form	0.0154	0.0147
CRR Tree ($M = 128$)	0.0154	0.0148
Monte Carlo ($M = 1048576$)	0.0154	0.0151

Table 6: Price Comparison between European and American Barrier Options

1.6.2 Vega Sensitivity Confront

Exploiting the linearity of the derivative operator, as already discussed in 1.5.1, we obtain a closed-form decomposition for the Vega of the American Up-and-Out option. In particular, consistently with the pricing representation introduced above, the total Vega can be written as the difference between the Vega of the terminal barrier component and that of the reflected term:

$$\begin{aligned}\nu_{U\&O} =& \nu_{\text{Call}}(F(0,T),K) - \nu_{\text{Call}}(F(0,T),KO) - \nu_{\text{Digital}}(F(0,T),KO) \\ &- \frac{F(0,T)}{KO}\Big[\nu_{\text{Call}}(F^{\text{ref}}(0,T),K) - \nu_{\text{Call}}(F^{\text{ref}}(0,T),KO) - \nu_{\text{Digital}}(F^{\text{ref}}(0,T),KO)\Big],\end{aligned} \tag{10}$$

where the reflected forward level is defined by

$$F^{\text{ref}}(0,T) = \frac{KO^2}{F(0,T)}. \tag{11}$$

As for the numerical estimates, we apply the same bump-and-revalue methodology described in 1.5.2, now using the pricing functions for the American Up-and-Out Barrier Call under both the Monte Carlo and the CRR frameworks. More precisely, the option is re-priced at the perturbed volatility levels $\sigma + h$ and $\sigma - h$, and Vega is then approximated through the corresponding central-difference estimator.
The results are summarized in Table 7.

Numerical Method	European Barrier	American Barrier
Exact Closed-Form	0.0014	0.0012
CRR Tree ($M = 128$)	0.0017	0.0010
Monte Carlo ($M = 1048576$)	0.0014	0.0013

Table 7: Vega Comparison between European and American Barrier Options

1.6.3 Delta Sensitivity Confront

The Delta (Δ) measures the sensitivity of the option price to changes in the underlying spot price S_0.

Methodology For the European barrier, we compared two numerical differentiation approaches for the estimation of Delta, consistently with the forward-based notation adopted above:

- **Direct Spot Bumping (Mode 1):** Delta is estimated by directly perturbing the initial spot price S_0 and re-pricing the option through the corresponding forward level $F(0,T)$:

$$\Delta \approx \frac{V(S_0+h)-V(S_0-h)}{2h}.$$

- **Forward Bumping and Chain Rule (Mode 2):** Delta is first computed with respect to the initial forward price $F(0,T)$ and then mapped back to the spot through the chain rule:

$$\Delta = \frac{\partial V}{\partial F(0,T)} \frac{\partial F(0,T)}{\partial S_0}.$$

Since

$$F(0,T) = S_0 \, e^{(r-q)T},$$

it follows that

$$\Delta = \frac{\partial V}{\partial F(0,T)} \, e^{(r-q)T}.$$

For the American barrier, the numerical estimates are obtained through direct Spot Bumping, while the exact Delta is derived analytically by differentiating the pricing formula based on the reflection term.
The results are summarized in Table 8.

	European Barrier (€)		American Barrier (€)
Numerical Method	**Spot Bump**	**Chain Rule**	**Spot Bump**
Exact Closed-Form	0.2160	0.2160	0.2241
CRR Tree ($M = 128$)	0.2346	0.2346	0.2185
Monte Carlo ($M = 1048576$)	0.2156	0.2156	0.2096

Table 8: Delta Comparison between European and American Barrier Options.

Discussion of Results The analysis of Table 8 highlights several relevant technical considerations:

- **CRR Model Stability:** In the binomial tree model, both calculation methods (Spot Bump and Chain Rule) yield identical results (0.2346). This is due to the discrete grid structure of the tree; as long as the "bump" h is not large enough to move terminal nodes across the barrier asymmetrically, the sensitivity remains constant. However, a slight discrepancy is noted compared to the exact value (0.2160), which is a typical result of the *barrier alignment problem*, that we have already mentioned above in 1.5.2.
- **Monte Carlo Precision:** The MC simulations show excellent convergence toward the analytical value for the European barrier. Moreover, again both methods show the same result (0.2156).
- **Impact of the American Barrier:** The exact Delta for the American barrier option is higher than that of the European counterpart (0.2241 vs 0.2160). This indicates that continuous monitoring makes the contract more sensitive to S_0 fluctuations: an increase in price not only raises the Call's intrinsic value but also drastically alters the probability of early knock-out throughout the entire life of the product, whereas the European barrier only "risks" termination at maturity.

1.7. Variance Reduction: Antithetic Variable Technique

To improve the efficiency of this estimator, we implemented the *antithetic variates* technique, a classical variance-reduction method based on introducing negative dependence between paired replications. In the case of simulations driven by standard normal random variables, this is achieved by pairing each draw $Z \sim \mathcal{N}(0,1)$ with its reflection $-Z$, which has the same distribution. Thus, instead of using

$2M$ independent normal draws, we construct M antithetic pairs and average the two corresponding payoffs. This preserves unbiasedness while potentially reducing variance, provided that the dependence induced between the paired outputs is sufficiently negative. In particular, the method is especially effective when the payoff is approximately monotone, or more generally close to a linear function of the simulated inputs.
In our MATLAB implementation, each simulation step is based on one standard normal draw Z and its antithetic counterpart $-Z$. This produces the following pair of terminal forward prices:

$$\begin{aligned} F(T,T)^{(1)} &= F(0,T)\exp\left(-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z\right), \\ F(T,T)^{(2)} &= F(0,T)\exp\left(-\frac{1}{2}\sigma^2 T - \sigma\sqrt{T}\,Z\right), \end{aligned} \qquad Z \sim \mathcal{N}(0,1). \tag{12}$$

Numerical Results

We compare the estimated standard error of the standard Monte Carlo ($\hat{\sigma}$) against the antithetic version ($\hat{\sigma}AV$) using the same number of simulation trials MMC. The ratio obtained from the execution is:

$$\frac{\hat{\sigma}}{\hat{\sigma}_{AV}} = 1.5204 \tag{13}$$

This result indicates a significant reduction in numerical noise; specifically, the standard error is reduced by approximately **34%**.

Observation

The high efficiency of the antithetic technique in this case is strictly linked to the monotonicity of the European Call option payoff with respect to the underlying price, which ensures that the antithetic pair generates a strong negative correlation between simulated results, since when one value is above the true value, the other tends to be below it. Consequently the errors in one path tend to cancel out the errors in its antithetic counterpart, leading to a much more stable and precise estimator.

1.8. h. Pricing of a Bermudan Call Option

To price the Bermudan call option, we use the CRR binomial tree framework. However, the backward induction procedure has been modified to satisfy the specific early exercise check required by the contract (in our case, at the end of each month). On these designated dates, we compare the continuation value, representing the discounted expected future payoffs, with the intrinsic value, which is the immediate gain from exercising the option ($S_t - K$). The option value at each node is then determined by the maximum of these two values.
Pricing the option with the given parameters, we get the following results:

Option Type	**Price (€)**
European Call	0.0163
Bermudan Call	0.0163

Table 9: Comparison between European and Bermudan option prices

Observation

A Bermudan option should always be priced higher than or equal to a European option ($C_{\text{Bermudan}} \geq C_{\text{European}}$) since the Bermudan structure gives the holder additional exercise opportunities (at the end of each month, in this case) before maturity.
In our case, we find that the Bermudan price and the European price are identical, and this can be explained by two factors:

- The option is currently Out-of-the-Money (OTM), as the underlying price ($S = 1.0$) is lower than the strike price ($K = 1.1$). Exercising early would result in an immediate loss, making the "continuation value" (holding the option) always superior to the "exercise value" in the binomial tree nodes.
- The risk-free rate ($r = 2.5\%$) is higher than the dividend yield ($d = 2\%$). By exercising early, the holder would gain the dividends but would lose the interest that could be earned by delaying the payment of the strike price K. Since $r > d$, it is optimal to delay the exercise until the final maturity to maximize the time value of money.

We can conclude that it is never optimal to early exercise this specific Bermudan call, and this is the reason why its price is identical to the price of a European call.

1.9. i. Dividend Yield Sensitivity: Comparing Bermudan and European Call Prices

We investigate the relationship between the dividend yield (d), ranging from 0% to 5%, and the prices of a Bermudan call option (computed via the CRR binomial model) and a European call option (computed via the Black-Scholes closed-form solution).

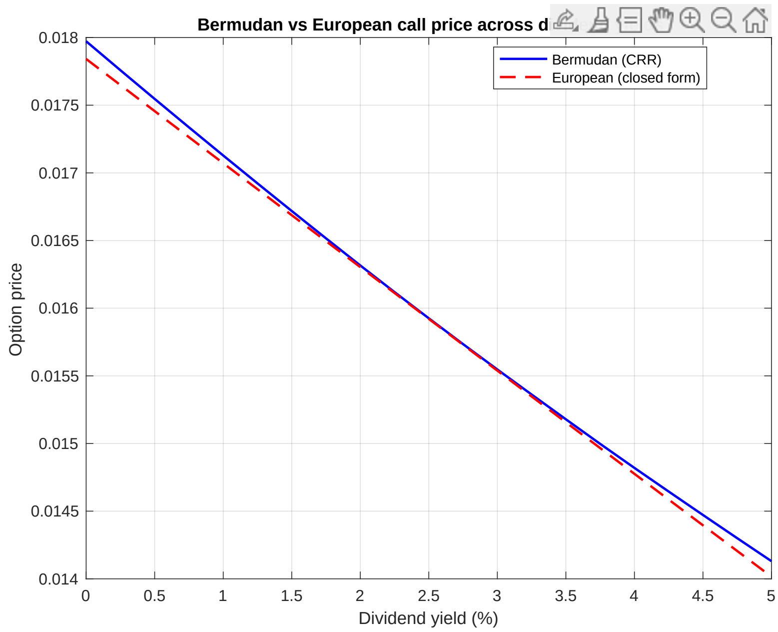

Figure 4: Comparison between Bermudan and European call prices as a function of dividend yield.

As expected, the prices of both Bermudan and European call options decrease as the dividend yield increases. This is because higher dividends reduce the expected future growth of the underlying stock price, lowering the value of call options.
Moreover, we observe that the gap between the Bermudan and European call prices widens as the dividend yield increases. This occurs because a higher dividend yield makes early exercise more convenient for the holder, who wants to get the dividend value before the stock price drops on the ex-dividend date.
On the other hand, we would expect that, when the dividend yield is zero or very low, the Bermudan and the European price should be identical. However, the plot reveals a slight discrepancy even at $d = 0\%$; this is a numerical discretization error since the CRR model is a discrete-time approximation of the underlying stochastic process, while the Black-Scholes model is a continuous-time closed-form solution.
In order to minimize this error and obtain higher precision, we can increase the number of time steps (M_{CRR}) in the binomial tree to ensure better convergence toward the continuous-time price.

1.10. WHAT IF the forward can assume negative values?

In standard Equity models like Black '76, the forward dynamics follow a Geometric Brownian Motion (GBM), which prevents negative values. If we assume the forward can assume negative values, we must adopt another model and, in particular, we choose an Arithmetic Brownian Motion (ABM), enforcing the zero drift condition to preserve martingality $E_{t_0}[F(t,t)] = F(0,t)$, which satisfies the no-Arbitrage condition for forwards.
Therefore we define the forward dynamics under the Risk-Neutral measure as:

$$dF(s,t) = \sigma dW_s, \quad F(t_0,t) = F_0 \tag{14}$$

where σ is the absolute volatility (expressed in currency units, not percentage).
Integrating from t_0 up to t:

$$F(t,t) = F_0 + \sigma(W_t - W_{t_0}) \tag{15}$$

At expiry, $F(t,t) \sim \mathcal{N}(F_0, \sigma^2(t-t_0))$.
The price of a European Call option $C(t_0,t)$ is the discounted expected value of the payoff:

$$C(t_0,t) = B(t_0,t)E_0[F(t,t) - K]^+ \tag{16}$$

Let $\tau = t - t_0$ and $Z \sim \mathcal{N}(0,1)$. We can write $F(t,t) = F_0 + \sigma\sqrt{\tau}Z$. The expectation is:

$$E_0[F(t,t) - K]^+ = \int_{-\infty}^{+\infty} \max(F_0 + \sigma\sqrt{\tau}z - K, 0)\phi(z)dz \tag{17}$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$ is the standard normal PDF. The payoff is positive only when $z > \frac{K-F_0}{\sigma\sqrt{\tau}}$.
Let, by a changing of variable, $d = \frac{F_0-K}{\sigma\sqrt{\tau}}$

$$E_0[F(t,t) - K]^+ = \int_{-d}^{+\infty} (F_0 - K + \sigma\sqrt{\tau}z)\phi(z)dz \tag{18}$$

Splitting the integral:

1. $(F_0 - K)\int_{-d}^{+\infty} \phi(z)dz = (F_0 - K)N(d)$
2. $\sigma\sqrt{\tau}\int_{-d}^{+\infty} z\phi(z)dz = \sigma\sqrt{\tau}\left[-\phi(z)\right]_{-d}^{+\infty} = \sigma\sqrt{\tau}\phi(d)$ where N is the standard normal CDF and ϕ is the standard normal PDF.

Combining the terms and applying the discount factor $B(t_0,t)$

$$C(t_0,t) = B(t_0,t)\left[(F_0 - K)N(d) + \sigma\sqrt{\tau}\phi(d)\right] \tag{19}$$

This formula allows for $F_0 < 0$ and provides a valid price even if the forward is negative. This model is known as Bachelier model.

1.11. Difference between an ATM forward Call and an ATM forward Put for a Market Maker

To evaluate the difference between an At-The-Money (ATM) forward call and an ATM forward put, we must examine the risk profile of a delta-hedged portfolio. A Market Maker (MM) does not view options in isolation; rather, they manage the aggregate Greeks of the hedged position.

Portfolio Composition and Delta Neutrality

Consider two portfolios, Π_C and Π_P, where the MM is long one option contract and delta-hedged using forward contracts (F). Given the ATM forward condition $K = F0$, and assuming no dividends ($q = 0$):

$$\Pi_C = C - \Delta_C F \tag{20}$$
$$\Pi_P = P - \Delta_P F \tag{21}$$

Using the Black '76 model, the forward delta Δ_F for an ATM forward option is:

$$\Delta_C = e^{-rT} N(d_1) \approx 0.5e^{-rT}, \quad \Delta_P = e^{-rT}[N(d_1) - 1] \approx -0.5e^{-rT} \tag{22}$$

By maintaining these hedge ratios, both portfolios achieve $\Delta_\Pi = 0$.

Equivalence of Greeks

We notice that the higher-order Greeks, which determine the portfolio's sensitivity to volatility and time, are mathematically identical:

- **Gamma (Γ):** $\Gamma_C = \Gamma_P = e^{-rT} \frac{N'(d_1)}{F\sigma\sqrt{T}}$. Both portfolios profit equally from realized price moves in either direction.
- **Vega (ν):** $\nu_C = \nu_P = e^{-rT} F\sqrt{T}N'(d_1)$. Both portfolios gain equally from an increase in implied volatility.
- **Theta (Θ):** $\Theta_{\Pi_C} = \Theta_{\Pi_P} = -\frac{S\sigma N'(d_1)}{2\sqrt{T}}$. The time decay paid to maintain the convexity is identical for both hedged positions.

The Synthetic Straddle and Payoff Profile

The equivalence is best demonstrated via Put-Call Parity: $C - P = e^{-rT}(F0 - K)$. At $K = F0$, $C = P$. Substituting this into the portfolio equations reveals:

$$\Pi_C \approx \Pi_P \approx \frac{1}{2}(\text{Straddle}) \tag{23}$$

where a **Straddle** is a strategy consisting in a long position in a Put Option and a long position in a Call Option. This confirms that for a delta-hedged MM, being long an ATM forward call is synthetically identical to being long an ATM forward put. Both positions result in a "Synthetic Straddle" profile, centered at the forward price.
Since an ATM Call and Put have the same price and same sensitivity at the forward strike ($C = P$), hedging either one results in an identical "V-shaped" payoff profile. This profile is equivalent to holding **half of a straddle**, as the hedged position provides the same volatility exposure without the directional risk.

Conclusion

For a Market Maker who maintains continuous delta-neutrality, there is **no functional difference** between an ATM forward call and an ATM forward put. The distinction is purely nominal. Whether the MM buys a call or a put, they are essentially "buying Gamma". This means they are paying a daily "rent" (Theta) in hopes that the underlying asset will fluctuate significantly, which will increase the delta of the portfolio (it becomes more positive), so the MM has to sell high some stock in order to re-hedge.
The ultimate profit or loss (P&L) does not depend on whether the market goes up or down. Instead, it is determined by the gap between **Realized Volatility** (actual market movement) and **Implied Volatility** (the price paid for the option). If the market moves more than the option price suggested, the Market Maker profits; if it moves less, the Market Maker loses money due to time decay.
When the underlying asset pays a continuous dividend yield q different from 0, the fundamental "no-difference" conclusion for the Market Maker remains the same, it only changes the mechanism of how they maintain that position.

1. Simulation of the default time

The default time τ is simulated by inverse transform sampling. Let $U \sim \mathcal{U}(0,1)$ and let

$$F(t) = 1 - S(t)$$

denote the distribution function associated with the prescribed survival probability. Then τ is obtained by solving

$$F(\tau) = U.$$

Equivalently, one may write $S(\tau) = 1 - U$. Since $1 - U$ has the same distribution as U, it is convenient to introduce

$$E = -\log U.$$

This yields the explicit inverse-transform representation

$$\tau = \begin{cases} E/\lambda_1, & E \leq \lambda_1\theta, \\ \theta + \dfrac{E - \lambda_1\theta}{\lambda_2}, & E > \lambda_1\theta. \end{cases}$$

2. Maximum-likelihood estimation of the survival model

Let

$$\boldsymbol{\tau} = \{\tau_i\}_{i=1}^{M}$$

denote the simulated sample. The parameters λ_1 and λ_2 are estimated by maximum likelihood, whereas θ is treated as known and fixed. To compute the density of a single observation, and then the likelihood of the dataset, we first verify the survival function, namely is differentiable with respect to the parameters of interest.

Indeed, for fixed θ, the survival function is differentiable with respect to λ_1 and λ_2 on each of the two regions $t < \theta$ and $t > \theta$. More precisely,

$$\frac{\partial S(t)}{\partial \lambda_1} = \begin{cases} -t\,S(t), & t < \theta, \\ -\theta\,S(t), & t > \theta, \end{cases} \qquad \frac{\partial S(t)}{\partial \lambda_2} = \begin{cases} 0, & t < \theta, \\ -(t-\theta)\,S(t), & t > \theta. \end{cases}$$

It remains to examine the behaviour at the cutoff point $t = \theta$. A direct computation shows that the one-sided derivatives coincide:

$$\left.\frac{\partial S(t)}{\partial \lambda_1}\right|_{t=\theta^-} = \left.\frac{\partial S(t)}{\partial \lambda_1}\right|_{t=\theta^+} = -\theta e^{-\lambda_1\theta}, \qquad \left.\frac{\partial S(t)}{\partial \lambda_2}\right|_{t=\theta^-} = \left.\frac{\partial S(t)}{\partial \lambda_2}\right|_{t=\theta^+} = 0.$$

Therefore, the derivatives with respect to λ_1 and λ_2 exist, and are continuous, at $t = \theta$. It follows that, for fixed θ, the survival function is differentiable on the whole domain $[0, +\infty)$ with continuous derivatives with respect to the parameter vector (λ_1, λ_2).

The density associated with the survival function is

$$f(t; \lambda_1, \lambda_2) = \begin{cases} \lambda_1 e^{-\lambda_1 t}, & t \leq \theta, \\ \lambda_2 e^{-\lambda_1\theta - \lambda_2(t-\theta)}, & t > \theta. \end{cases}$$

Assuming that the sample is i.i.d., the likelihood function is the product of the individual densities:

$$L(\lambda_1, \lambda_2; \boldsymbol{\tau}, \theta) = \prod_{i=1}^{M} f(\tau_i; \lambda_1, \lambda_2).$$

Define the index sets

$$I_1 = \{i : \tau_i \leq \theta\}, \qquad I_2 = \{i : \tau_i > \theta\},$$

and let $N_1 = |I_1|$, $N_2 = |I_2|$, with $N_1 + N_2 = M$. Then the likelihood can be written as

$$L(\lambda_1, \lambda_2; \boldsymbol{\tau}, \theta) = \prod_{i \in I_1} \lambda_1 e^{-\lambda_1 \tau_i} \prod_{i \in I_2} \lambda_2 e^{-\lambda_1 \theta - \lambda_2 (\tau_i - \theta)}.$$

The log-likelihood is therefore

$$\ell(\lambda_1, \lambda_2) = N_1 \log \lambda_1 + N_2 \log \lambda_2 - \lambda_1 \left(\sum_{i \in I_1} \tau_i + N_2 \theta \right) - \lambda_2 \sum_{i \in I_2} (\tau_i - \theta).$$

The first-order conditions yield the closed-form estimators

$$\widehat{\lambda}_1 = \frac{N_1}{\sum_{i \in I_1} \tau_i + N_2 \theta}, \qquad \widehat{\lambda}_2 = \frac{N_2}{\sum_{i \in I_2} (\tau_i - \theta)}.$$

Moreover,

$$\frac{\partial^2 \ell}{\partial \lambda_1^2} = -\frac{N_1}{\lambda_1^2} < 0, \qquad \frac{\partial^2 \ell}{\partial \lambda_2^2} = -\frac{N_2}{\lambda_2^2} < 0, \qquad \frac{\partial^2 \ell}{\partial \lambda_1 \partial \lambda_2} = 0,$$

hence the log-likelihood is concave and the critical point is the unique global maximizer.
To compute confidence intervals, we briefly recall a standard result on the asymptotic Gaussianity of the maximum-likelihood estimator. Let $\theta \in \mathbb{R}$ be a parameter to estimate and $\hat{\theta}_{\mathrm{MLE}}$ denote its maximum-likelihood estimator. If

the observations are i.i.d.;

the model is identifiable;

the support of the density is common and does not depend on the parameter;

the true parameter belongs to the interior of the parameter space;

the density is sufficiently differentiable with respect to the parameter, and the required interchange between differentiation and integration is valid.

the maximum-likelihood estimator is asymptotically Gaussian, i.e.

$$\sqrt{M}(\hat{\theta}_{\mathrm{MLE}} - \theta) \xrightarrow{d} \mathcal{N}\big(0, I(\theta)^{-1}\big),$$

where $I(\theta)$ is the Fisher information for one observation. For a multidimensional parameter $\boldsymbol{\theta} \in \mathbb{R}^p$ with maximum likelihood estimator $\hat{\boldsymbol{\theta}}_{\mathrm{MLE}}$ it holds:

$$\sqrt{M}(\hat{\boldsymbol{\theta}}_{\mathrm{MLE}} - \boldsymbol{\theta}) \xrightarrow{d} \mathcal{N}_p\big(\mathbf{0}, I(\boldsymbol{\theta})^{-1}\big).$$

In our setting, the parameter vector is given by

$$\boldsymbol{\lambda} = \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix}, \qquad \hat{\boldsymbol{\lambda}} = \begin{pmatrix} \widehat{\lambda}_1 \\ \widehat{\lambda}_2 \end{pmatrix}.$$

Since the observations form an i.i.d. sample and the associated parametric model belongs to the exponential family, the regularity conditions established above are satisfied, therefore the standard asymptotic theory for maximum-likelihood estimation applies.

$$\sqrt{M}\big(\hat{\boldsymbol{\lambda}} - \boldsymbol{\lambda}\big) \xrightarrow{d} \mathcal{N}_2\big(\mathbf{0}, I(\boldsymbol{\lambda})^{-1}\big).$$

Accordingly, $\hat{\boldsymbol{\lambda}}$ is asymptotically bivariate normal, with asymptotic covariance matrix given by the inverse of the Fisher information matrix.

Observed information, asymptotic covariance matrix, confidence regions, and numerical results

The observed information matrix evaluated at the maximum-likelihood estimator is

$$\mathrm{Info}(\hat{\lambda}_1, \hat{\lambda}_2) = \begin{pmatrix} N_1/\hat{\lambda}_1^2 & 0 \\ 0 & N_2/\hat{\lambda}_2^2 \end{pmatrix}.$$

Its inverse provides the asymptotic covariance matrix of the estimator,

$$\mathrm{Cov} = \begin{pmatrix} \hat{\lambda}_1^2/N_1 & 0 \\ 0 & \hat{\lambda}_2^2/N_2 \end{pmatrix}.$$

Accordingly, the marginal standard errors are

$$\mathrm{SE}(\hat{\lambda}_1) = \frac{\hat{\lambda}_1}{\sqrt{N_1}}, \qquad \mathrm{SE}(\hat{\lambda}_2) = \frac{\hat{\lambda}_2}{\sqrt{N_2}}.$$

For a significance level α, the corresponding marginal confidence intervals are

$$\hat{\lambda}_j \pm z_{1-\alpha/2}\,\mathrm{SE}(\hat{\lambda}_j), \qquad j = 1, 2.$$

The joint confidence region for the parameter vector is given by the ellipse

$$\left\{\boldsymbol{x} \in \mathbb{R}^2 : (\boldsymbol{x} - \hat{\boldsymbol{\lambda}})^\top \mathrm{Cov}^{-1}(\boldsymbol{x} - \hat{\boldsymbol{\lambda}}) \leq \chi^2_{2,1-\alpha}\right\}.$$

In the numerical experiment, based on a sample of size $M = 10^5$, the maximum-likelihood estimates are

$$\hat{\lambda}_1 = 4.14 \times 10^{-4}, \qquad \hat{\lambda}_2 = 1.00 \times 10^{-3}.$$

The corresponding marginal confidence intervals are

$$\mathrm{CI}(\lambda_1) = \left[3.58 \times 10^{-4},\, 4.71 \times 10^{-4}\right], \qquad \mathrm{CI}(\lambda_2) = \left[9.94 \times 10^{-4},\, 1.006 \times 10^{-3}\right].$$

These results are fully consistent with the theoretical parameter values used in the simulation, namely

$$\lambda_1 = 4 \times 10^{-4}, \qquad \lambda_2 = 10 \times 10^{-4} = 10^{-3},$$

since both true values lie within the corresponding confidence intervals. Moreover, the estimate of λ_2 is extremely accurate, while the interval associated with λ_1 is comparatively wider. This behaviour is consistent with the structure of the model: the first regime, corresponding to times before the cutoff θ, contains less effective information on λ_1, whereas the second regime provides a larger amount of information on λ_2.

The graphical analysis confirms the same conclusion. The empirical survival probability and the fitted survival curve are nearly indistinguishable on the loglinear plot, and the change in slope at $t = \theta$ is correctly reproduced by the fitted model. In addition, the joint confidence region has the expected elliptical shape induced by the asymptotic Gaussianity of the maximum-likelihood estimator.

3. Response to the assignment questions

(a) Simulation of the default time. The default time is simulated exactly by inverse transform sampling, using the explicit piecewise representation

$$\tau = \begin{cases} E/\lambda_1, & E \leq \lambda_1\theta, \\ \theta + \dfrac{E - \lambda_1\theta}{\lambda_2}, & E > \lambda_1\theta, \end{cases} \qquad E = -\log U, \quad U \sim \mathcal{U}(0,1).$$

This procedure generates an i.i.d. sample of size $M = 10^5$ from the default-time distribution prescribed in the statement.

(b) Fit of the survival probability. The fit is performed by maximum likelihood, with θ treated as known and fixed, and with only λ_1 and λ_2 being estimated. The fitted survival function is therefore

$$\widehat{S}(t) = \begin{cases} e^{-\hat{\lambda}_1 t}, & t \leq \theta, \\ e^{-\hat{\lambda}_1 \theta - \hat{\lambda}_2 (t-\theta)}, & t > \theta. \end{cases}$$

The fitted values are

$$\hat{\lambda}_1 = 4.14 \times 10^{-4}, \qquad \hat{\lambda}_2 = 1.00 \times 10^{-3},$$

which are in close agreement with the true parameters. The goodness of fit is assessed by comparing the experimental survival probability with the fitted survival probability on a loglinear plot, where the two curves are seen to be almost superimposed. Finally, parameter uncertainty is quantified both through the marginal confidence intervals

$$\mathrm{CI}(\lambda_1) = \left[3.58 \times 10^{-4},\, 4.71 \times 10^{-4}\right], \qquad \mathrm{CI}(\lambda_2) = \left[9.94 \times 10^{-4},\, 1.006 \times 10^{-3}\right],$$

and through the joint confidence ellipse for the vector parameter (λ_1, λ_2). Overall, the numerical evidence shows that the estimation procedure successfully recovers the true parameters of the simulated survival model.